Arithmetic Quantum Cat Maps:
Exact Fourth-Moment Reductions, Direct Cayley Energy Bounds, and the Remaining BGS Frontier

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Abstract

We develop an exact finite-field reduction of a fourth-moment problem arising in arithmetic quantum cat maps and isolate the arithmetic obstruction that remains before any microscopic Bohigas–Giannoni–Schmit (BGS) statement can be contemplated. The pair-key geometry gives an exact constrained-pair count $(K-1)(K-2)$ and a corrected fourth-moment difference distribution

$$B(t) = O(t) + 2(p-2)C_H(t) + (p-2)^2\mathbf{1}_{t=0}.$$

A Cayley transform turns the mixed term into the nonzero additive-difference energy of a Möbius image $A = c(H \setminus \{1\})$ of a multiplicative subgroup. For odd prime subgroup order K, Chowla's rational independence theorem for cotangents gives a characteristic-zero classification of all equal-difference collisions, and every cleared collision polynomial satisfies

$$\mathrm{Res}(\Phi_K, P) = K^3M^2, \qquad M \in \mathbf{Z}.$$

An explicit slice has $\mathrm{Res}(\Phi_K, X^2+3X+1) = F_K^2$, so the square factor can be exponentially large. Nevertheless its rational Cayley model has exact nonzero difference energy $2N(4N-3)+32$, and a 28-digit finite-field certificate reproduces the same excess 32 at $K = 131$.

The mixed term can also be controlled directly, without excluding exceptional primes. For every subgroup of order

$$100 < K < \frac{1}{3}p^{3/4},$$

a source-exact Stepanov estimate on an absolutely irreducible bilinear fiber gives

$$E_+(A) < 34K^{8/3}.$$

For odd prime K we further prove a global split-prime budget: if $\Delta_K(p)$ denotes the non-forced mixed-energy excess, then

$$\sum_{p\equiv 1\,(K)} \Delta_K(p)\log p \le (K^3-5K^2+6K+1)\big((K-1)\log 12 - 3\log K\big).$$

Thus a polynomial-scale pointwise mixed theorem and an $O(K^4)$ global exceptional-mass theorem are available. The sharper centered target $E_+(A) = (K-1)^4/p + O(K^{2+o(1)})$ remains open. The off/off elliptic-family estimate remains a theorem draft, and nothing here proves microscopic BGS/GOE eigenphase statistics.

1 Scope and evidence discipline

Throughout, p is an odd prime and $H \le \mathbf{F}_p^\times$ is a multiplicative subgroup of order K. Division is in $\mathbf{F}_p$. We distinguish:

- **Exact**: a current proof with explicit hypotheses;
- **Finite certificate**: an exhaustive or kernel-checked statement at explicit finite parameters;
- **Finite regression**: an implementation check that supports, but does not prove, a universal statement;
- **Numerical**: descriptive floating-point evidence only;
- **Draft/Open**: a named closure condition remains.

The distinction between moment information and microscopic spectral statistics is essential. Arithmetic cat-map literature contains strong matrix-element, eigenfunction and quantum-ergodicity results [1, 2, 3, 4, 5], but these are not interchangeable with a local eigenphase law. Conversely, rigorous BGS-type results are known for special model classes such as mixing quantum graphs under their own large-graph hypotheses [6, 7]; they do not transfer to the arithmetic family treated here without a new argument.

2 Exact pair-key geometry

For $x, y \in \mathbf{F}_p \setminus \{0, 1\}$ define

$$u = \frac{x}{y}, \qquad v = \frac{x-1}{y-1}, \qquad d = x - y. \tag{1}$$

Proposition 2.1 (Off-diagonal pair-key bijection)**.** *If $x \neq y$, then $u, v \in \mathbf{F}_p^\times \setminus \{1\}$, $u \neq v$, and*

$$y = \frac{1-v}{u-v}, \qquad x = \frac{u(1-v)}{u-v}, \qquad d = \frac{(u-1)(v-1)}{v-u}. \tag{2}$$

Conversely every such ordered pair (u, v) recovers a unique admissible off-diagonal (x, y). The diagonal $x = y$ collapses to $(u, v, d) = (1, 1, 0)$ with multiplicity $p - 2$.

Proof. From $x = uy$ and $x - 1 = v(y - 1)$, $(u - v)y = 1 - v$, which gives the inverse formulas and then the formula for d. The diagonal statement is immediate. □

Corollary 2.2 (Exact constrained-pair count)**.** *If the two constraints are $u, v \in H$, the number of admissible off-diagonal ordered pairs is*

$$(K - 1)(K - 2).$$

Proof. Choose an ordered pair of distinct elements of $H \setminus \{1\}$ and apply Proposition 2.1. □

Status: Exact.. Independent Python/C++ checks and a Lean certificate at $(p, K) = (53, 13)$ guard the implementation, but are not the source of the general proof.

3 Fourth-moment coupling and finite Fourier identities

Give each collapsed key $r = (u_r, v_r, d_r)$ multiplicity n_r, equal to one off the diagonal and $p - 2$ at $(1, 1, 0)$. Define

$$B(t) = \sum_{r,s} n_r n_s \, \mathbf{1}_H(u_r u_s) \mathbf{1}_H(v_r v_s) \mathbf{1}_{d_r + d_s = t}. \tag{3}$$

Let $O(t)$ be the off/off contribution and

$$C_H(t) = \#\left\{(u,v) \in (H \setminus \{1\})^2 : u \neq v,\ \frac{(u-1)(v-1)}{v-u} = t\right\}.$$

Then the diagonal bookkeeping is exactly

$$B(t) = O(t) + 2(p-2)C_H(t) + (p-2)^2 \mathbf{1}_{t=0}. \tag{4}$$

The coefficient 2 is load-bearing: diagonal/off and off/diagonal are distinct ordered classes.

Proposition 3.1 (Exact total coupled count)**.**

$$|V_4| = (K^2+1)p^2 - (6K^2+3K)p + 13K^2 - 3K.$$

Let $S = H^\perp$ be the multiplicative-character annihilator, so $m = |S| = (p-1)/K$. In the maintained twisted-Jacobi convention,

$$\sum_{\chi,\psi \in S} |J(\chi,\psi;q)|^4 = m^2 \Sigma_4(q), \qquad \Sigma_4(q) = \sum_{t \in \mathbf{F}_p} B(t) e_p(qt). \tag{5}$$

Writing $V = \sum_t B(t)$ and $Z = B(0)$,

$$\mathbb{E}_{q \neq 0} \Sigma_4(q) = \frac{pZ - V}{p-1}, \qquad \mathrm{Var}_{q \neq 0}\, \Sigma_4(q) = p\, \mathrm{Var}_{t \neq 0}\, B(t). \tag{6}$$

These follow directly from finite Fourier orthogonality and Parseval; no independence heuristic enters.

4 Cayley transform and additive energy

Put

$$c(h) = \frac{h+1}{h-1}, \qquad h \in H \setminus \{1\}.$$

For $u \neq v$,

$$\frac{(u-1)(v-1)}{v-u} = \frac{2}{c(u) - c(v)}. \tag{7}$$

Thus for $A = c(H \setminus \{1\})$ the nonzero multiplicities of C_H are the nonzero difference multiplicities of A, up to the permutation $x \mapsto 2/x$. Consequently

$$\sum_t C_H(t)^2 = E_+(A) - |A|^2, \qquad E_+(A) = \sum_x r_{A-A}(x)^2. \tag{8}$$

Since $c(h^{-1}) = -c(h)$, $A = -A$ and $0 \notin A$. With $n = |A| = K-1$, identity and inversion-swap alone force the nonzero-energy baseline

$$n(2n-3) = (K-1)(2K-5). \tag{9}$$

A separate exact matching audit of all $6! = 720$ integer systems obtained by clearing and matching Cayley collisions gives ranks

$$30 \text{ (rank 2)}, \qquad 240 \text{ (rank 3)}, \qquad 450 \text{ (rank 4)},$$

and an extra rank drop can occur only in characteristic 2. Complete fixed-K cyclotomic enumeration currently gives

$$E_{11} = \{23, 67, 89\}, \qquad E_{13} = \{53, 79, 131, 157, 313\}. \tag{10}$$

These are exact computer-assisted finite statements, not growing-K theorems.

5 Prime-order characteristic-zero rigidity

Let K now be an odd prime and $\zeta = e^{2\pi i/K}$. For $1 \le j \le K-1$,

$$c(\zeta^j) = -i \cot \frac{\pi j}{K}. \tag{11}$$

Chowla proved that

$$\cot \frac{\pi a}{K}, \qquad 1 \le a \le \frac{K-1}{2},$$

are linearly independent over $\mathbf{Q}$ [8]; Wang states this prime case explicitly and extends it [9].

Lemma 5.1 (Cross-polytope equal differences)**.** *Let x, y, u, v be vertices of $\{\pm e_1, \dots, \pm e_m\}$ with $x \neq y$ and $u \neq v$. If $x - y = u - v$, then either $(u, v) = (x, y)$ or $(u, v) = (-y, -x)$.*

Proof. The equality is $x + v = u + y$. If the common sum vanishes, then $v = -x$ and $u = -y$. Otherwise a nonzero sum of two signed basis vectors uniquely determines its unordered pair of summands. The alternative pairing would force $x = y$, excluded. □

Theorem 5.2 (Prime-order collision rigidity)**.** *Every characteristic-zero equal-difference collision in $\{c(\zeta^j) : 1 \le j < K\}$ is either identity or central inversion-swap.*

Proof. Use (11), replace each cotangent with the corresponding signed basis vector via Chowla's rational independence, and apply Lemma 5.1. □

6 Square cyclotomic resultants

For nonzero residues $i, j, r, s \pmod K$, with $i \neq j$ and $r \neq s$, choose integer lifts and define the cleared collision polynomial

$$\begin{aligned} P(X) = {} & X^{r+s+i} - X^{r+s+j} - X^{r+i+j} + 2X^{r+j} - X^r \\ & + X^{s+i+j} - 2X^{s+i} + X^s + X^i - X^j. \end{aligned} \tag{12}$$

Direct denominator clearing gives

$$2P(X) = \big(c(X^r) - c(X^s) - c(X^i) + c(X^j)\big) \prod_{a \in \{r,s,i,j\}} (X^a - 1). \tag{13}$$

The coefficient ℓ^1 norm satisfies $\|P\|_1 \le 12$.

Lemma 6.1 (Cubic zero and reciprocity)**.** *Writing $T = i + j + r + s$, one has*

$$P(X) = (X-1)^3 Q(X), \qquad Q \in \mathbf{Z}[X],$$

and

$$Q(X^{-1}) = X^{3-T} Q(X).$$

Proof. In (13), cancel the denominator of each Cayley term separately; every summand retains three factors $X^a - 1$, proving the cubic zero. Furthermore $c(X^{-a}) = -c(X^a)$ and $X^{-a} - 1 = -(X^a - 1)/X^a$. The four denominator factors contribute positive sign and total power X^{-T}, while the Cayley-difference contributes one minus sign. Hence $P(X^{-1}) = -X^{-T} P(X)$, which gives the formula for Q. □

Theorem 6.2 (Square-resultant theorem)**.** *For every odd prime K and every admissible collision quadruple,*

$$\operatorname{Res}(\Phi_K, P) = K^3 M^2 \tag{14}$$

for some $M \in \mathbf{Z}$. For a non-forced characteristic-zero collision, $M \neq 0$.

Proof. Let $L = \mathbf{Q}(\zeta)$ and L^+ be its maximal real subfield. Since 2 is invertible modulo K, choose m with $2m \equiv 3 - T \pmod{K}$ and put $\beta = \zeta^m Q(\zeta)$. By Lemma 6.1,

$$\overline{\beta} = \zeta^{-m} Q(\zeta^{-1}) = \zeta^{-m+3-T} Q(\zeta) = \beta.$$

Thus β is an algebraic integer in L^+. Multiplication by ζ^m does not change the norm, so

$$\mathrm{N}_{L/\mathbf{Q}}(Q(\zeta)) = \mathrm{N}_{L^+/\mathbf{Q}}(\beta)^2 = M^2.$$

Also $\mathrm{N}_{L/\mathbf{Q}}(\zeta - 1) = \Phi_K(1) = K$. Taking norms of $P(\zeta) = (\zeta - 1)^3 Q(\zeta)$ proves (14). The non-forced case has $P(\zeta) \neq 0$ by Theorem 5.2. □

Corollary 6.3 (Improved no-collision cutoff)**.** *Suppose $K \mid p - 1$. If a non-forced finite-field collision exists, then*

$$K^3 p^2 \leq 12^{K-1}. \tag{15}$$

Consequently $K^3 p^2 > 12^{K-1}$ implies the exact baseline (9).

Proof. A finite-field collision makes P and Φ_K share a root modulo p, so $p \mid \mathrm{Res}(\Phi_K, P)$. Since $K \mid p - 1$, $p \neq K$. Equation (14) gives $p \mid M$, hence $K^3 p^2 \leq |\mathrm{Res}(\Phi_K, P)|$. On the other hand $|P(\zeta^a)| \leq \|P\|_1 \leq 12$ for all a, giving the upper bound. □

The cutoff remains exponential in K. The next section shows that this is not merely an artifact of a crude norm estimate.

7 An explicit Fibonacci collision slice

Consider

$$c(h^2) - c(h) = c(h) - c(h^{-1}). \tag{16}$$

Since $c(h^{-1}) = -c(h)$,

$$c(h^2) - 3c(h) = -\frac{2(h^2 + 3h + 1)}{(h - 1)(h + 1)}.$$

For odd-order $h \neq 1$ we also have $h \neq -1$, so (16) is equivalent to

$$q(h) = h^2 + 3h + 1 = 0. \tag{17}$$

Theorem 7.1 (Fibonacci resultant identity)**.** *For every odd prime K,*

$$\mathrm{Res}(\Phi_K(X), X^2 + 3X + 1) = F_K^2. \tag{18}$$

For the corresponding cleared Cayley collision polynomial, the full resultant is $K^3 F_K^2$.

Proof. Let α, β be the roots of $X^2 + 3X + 1$, so $\alpha\beta = 1$, $\alpha + \beta = -3$, and $\Phi_K(X) = (X^K - 1)/(X - 1)$. Therefore

$$\mathrm{Res}(\Phi_K, q) = \frac{(\alpha^K - 1)(\beta^K - 1)}{(\alpha - 1)(\beta - 1)} = \frac{2 - (\alpha^K + \beta^K)}{5}.$$

With $\varphi = (1 + \sqrt{5})/2$, the roots are $-\varphi^2$ and $-\varphi^{-2}$. Since K is odd, $\alpha^K + \beta^K = -L_{2K}$, and the standard identity $L_{2K} = 5F_K^2 - 2$ for odd K gives (18). Comparing with Theorem 6.2, or directly taking the cleared denominator norm, gives $K^3 F_K^2$. □

Because $F_K \sim \varphi^K/\sqrt{5}$, Theorem 7.1 proves that no bound $|M| \leq K^C$ can hold for all collision resultants. Thus any useful polynomial-scale theorem must exploit eligible prime divisors or directly bound energy rather than total resultant size.

If p is prime and $K \mid p - 1$, then (16) occurs over $\mathbf{F}_p$ if and only if $p \mid F_K$. Indeed Φ_K splits completely over $\mathbf{F}_p$, and (18) is exactly the common-root criterion.

8 Fibonacci/Lucas ratios form an almost-Sidon set

Assume $q(h) = 0$ and put $t = c(h)$. Substitution $h = (t+1)/(t-1)$ into $q(h) = 0$ gives

$$5t^2 = 1. \tag{19}$$

A direct Binet calculation, equivalently an induction using the standard Fibonacci/Lucas recurrences, gives

$$F_n c(h^n) = L_n c(h), \qquad n \geq 1. \tag{20}$$

Thus the characteristic-zero model for half of the centrally symmetric Cayley set is

$$x_n = \frac{L_n}{F_n}, \qquad S_N = \{\pm x_1, \ldots, \pm x_N\}.$$

Two identities used repeatedly are

$$x_m + x_n = \frac{2F_{m+n}}{F_m F_n}, \qquad |x_m - x_n| = \frac{2F_{|m-n|}}{F_m F_n}. \tag{21}$$

Carmichael's primitive-divisor theorem implies that F_j has a prime divisor not dividing any earlier Fibonacci number for every $j > 12$ [10, 11]. We use this only as a divisibility separation principle; the finite range through 12 is checked exactly.

Theorem 8.1 (Exact almost-Sidon energy)**.** *For every $N \geq 3$, if*

$$E^*(S_N) = \sum_{d \neq 0} r_{S_N - S_N}(d)^2,$$

then

$$E^*(S_N) = 2N(4N-3) + 32. \tag{22}$$

The excess over the centrally symmetric forced baseline is exactly 32, independent of N.

Proof. First classify relations between positive x_n.

For equal absolute differences, suppose $m > n$, $r > s$ and

$$\frac{F_{m-n}}{F_m F_n} = \frac{F_{r-s}}{F_r F_s}.$$

If, say, $m > r$ and $m > 12$, a primitive divisor of F_m divides the right cross-multiplied side but none of F_{m-n}, F_r, F_s, a contradiction. Thus beyond the bounded range the upper indices agree. Then F_{m-n}/F_n is strictly decreasing in n, so the lower indices agree.

For equal sums, if $m+n > r+s$ and $m+n > 12$, a primitive divisor of F_{m+n} cannot divide any of F_{r+s}, F_m, F_n, again a contradiction. Thus the index sums agree. Assume $m \leq n$, $r \leq s$, and put $T = m+n = r+s$. Equality reduces to $F_m F_n = F_r F_s$. The exact product identity

$$5F_m F_n = L_T - (-1)^m L_{n-m}$$

then forces the unordered pairs $\{m, n\}$ and $\{r, s\}$ to agree.

Finally $1 \leq x_n \leq 3$. A positive sum can equal an absolute difference only at the endpoint value 2, giving $x_1 + x_1 = x_2 - x_1$. The remaining bounded relations are exactly those involving $x_1 = 1, x_2 = 3, x_3 = 2$:

$$|x_3 - x_1| = |x_2 - x_3| = 1, \quad x_1 + x_1 = x_2 - x_1 = 2, \quad x_1 + x_2 = x_3 + x_3 = 4.$$

The finite residual statement is independently kernel-checked in Lean for indices through 12.

For a centrally symmetric set of size $2N$, forced inversion pairing alone contributes $2N(4N-3)$. The two oriented bins ± 1 have multiplicity 4 rather than two independent multiplicity-2 bins, contributing excess 8 each. The four bins $\pm 2, \pm 4$ have multiplicity 3 rather than a multiplicity-1 and a multiplicity-2 bin, contributing excess 4 each. The total excess is $2 \cdot 8 + 4 \cdot 4 = 32$. □

Remark 8.2. *The broader theorem of Bilu–Hanrot–Voutier [12] supplies primitive divisors for much more general Lucas/Lehmer sequences. For Theorem 8.1, the classical Fibonacci theorem already suffices.*

9 A 28-digit exact finite-field certificate

Set

$$K = 131, \qquad p = F_{131} = 1066340417491710595814572169.$$

The repository proves the primality of p by the Lucas criterion using the complete factorization

$$p - 1 = 2^3 \cdot 11 \cdot 89 \cdot 131 \cdot 199 \cdot 521 \cdot 2081 \cdot 9901 \cdot 19801 \cdot 24571$$

and base 3: it checks

$$3^{p-1} \equiv 1 \pmod{p}, \qquad \gcd\bigl(3^{(p-1)/q} - 1, p\bigr) = 1$$

for every prime q in this factorization.

The explicit integer

$$h = 188706367382658195937453923$$

satisfies

$$h^2 + 3h + 1 \equiv 0 \pmod{p}, \qquad h^{131} \equiv 1 \pmod{p}, \qquad h \neq 1.$$

Since 131 is prime, h has exact order 131. Exact arbitrary-precision computation gives

$$\sum_t C_H(t)^2 = 33442, \qquad (K-1)(2K-5) = 33410, \qquad \text{excess} = 32. \tag{23}$$

The maximum nonzero difference multiplicity is 4, with histogram

$$1^{\times 126}, \qquad 2^{\times 8312}, \qquad 3^{\times 4}, \qquad 4^{\times 2}.$$

Independent Python and C++20 Boost.Multiprecision implementations verify these numbers and the primality certificate. The finite low-index relation classification in Theorem 8.1 is separately checked by Lean 4.

Status: Finite certificate.. This example is not extrapolated to an infinite sequence of Fibonacci primes. Its role is to show that very large exceptional split primes can coexist with only constant excess energy.

For comparison, the independent design $(p, K) = (461, 23)$ has

$$\sum_t C_H(t)^2 = 1190, \qquad \text{baseline} = 902, \qquad \text{excess} = 288.$$

Thus exceptional reductions can have different collision geometry.

10 Direct polynomial-scale Cayley energy

The Fibonacci obstruction changes the useful closure condition from "eliminate every extra collision" to "control total collision energy". In fact a uniform polynomial-scale theorem follows directly from the sum fibers of A.

For $t \in \mathbf{F}_p$, put

$$r(t) = \#\{(a, b) \in A^2 : a + b = t\}.$$

If $a = c(x)$ and $b = c(y)$ with $x, y \in H \setminus \{1\}$, clearing the two Cayley denominators gives

$$P_t(x, y) = (2 - t)xy + t(x + y) - (2 + t) = 0. \tag{24}$$

Lemma 10.1 (Absolute irreducibility of every sum fiber)**.** *For every $t \in \mathbf{F}_p$, the polynomial P_t is absolutely irreducible.*

Proof. A nonconstant bilinear polynomial $axy + bx + cy + d$ factors over the algebraic closure only if $ad - bc = 0$. For (24),

$$(2 - t)(-(2 + t)) - t^2 = -4 \neq 0$$

because p is odd. The linear special case $t = 2$ is also irreducible. □

Theorem 10.2 (Direct Cayley energy bound)**.** *Let p be odd and let $H \leq \mathbf{F}_p^\times$ have order K. If*

$$100 < K < \frac{1}{3}p^{3/4}, \tag{25}$$

then, for $A = c(H \setminus \{1\})$,

$$E_+(A) < 34K^{8/3}. \tag{26}$$

Consequently

$$\sum_t C_H(t)^2 < 34K^{8/3}. \tag{27}$$

Proof. For $t \neq 0, -2$, the polynomial (24) has bidegree $(1, 1)$,

$$P_t(0, 0) = -(2 + t) \neq 0, \qquad \deg P_t(x, 0) = 1.$$

Theorem 2 of Vyugin–Makarychev [13] applies to the subgroup product $H \times H$ under (25) and gives

$$r(t) \leq 32K^{2/3}.$$

The theorem actually counts the enlarged fiber in H^2, including the extraneous cleared-denominator point $(1, 1)$, so it is a valid upper bound for $r(t)$.

Since $A = -A$ and $0 \notin A$, $r(0) = K - 1$. Trivially $r(-2) \leq K - 1$. Using $\sum_t r(t) = |A|^2 = (K - 1)^2$,

$$\begin{aligned} E_+(A) &= \sum_t r(t)^2 \\ &\leq 2(K - 1)^2 + 32K^{2/3} \sum_{t \neq 0, -2} r(t) \\ &< 34K^{8/3}. \end{aligned}$$

Equation (27) follows from (8). □

The load-bearing source is the stated Theorem 2 of the 2015 arXiv version [13]; the later journal publication is [14]. In particular, the source condition is $\deg P(x, 0) \geq 1$, not a strict inequality. This was checked against the primary PDF because dropping the equality bar would make the present degree-one application invalid.

Let $M(t) = 2(p - 2)C_H(t)$ denote the mixed summand in (4). Since variance is bounded by the corresponding second moment,

$$\mathrm{Var}_{t \neq 0} M(t) \leq \frac{4(p - 2)^2}{p - 1} \sum_t C_H(t)^2 < 136pK^{8/3}. \tag{28}$$

Thus a genuine pointwise polynomial-scale mixed bound is proved. The exponent $8/3$, however, still loses $K^{2/3}$ relative to the natural quadratic collision scale.

11 Centered energy and the exact Fourier/character target

A raw global estimate $E_+(A) = O(K^2)$ cannot be the correct statement in every density regime. With the unnormalized additive Fourier transform

$$\widehat{1_A}(q) = \sum_{a \in A} e_p(qa),$$

Parseval gives the exact identity

$$E_+(A) = \frac{|A|^4}{p} + \frac{1}{p} \sum_{q \neq 0} |\widehat{1_A}(q)|^4. \tag{29}$$

The zero-frequency density term is unavoidable. The sharp pointwise target is therefore

$$E_+(A) = \frac{(K-1)^4}{p} + O(K^{2+o(1)}), \tag{30}$$

or equivalently

$$\sum_{q \neq 0} |\widehat{1_A}(q)|^4 \ll pK^{2+o(1)}.$$

In the intended regime $K = o(p^{1/4})$, the density term is negligible at the quadratic collision scale.

There is an exact character decomposition. Put $S = H^\perp$ and $m = |S| = (p-1)/K$. With $y = (h-1)^{-1}$,

$$h = \frac{y+1}{y}, \qquad c(h) = 1 + 2y,$$

and character orthogonality gives

$$\widehat{1_A}(q) = \frac{e_p(q)}{m} \sum_{\chi \in S} T_\chi(q), \qquad T_\chi(q) = \sum_{y \neq 0, -1} \chi\left(\frac{y+1}{y}\right) e_p(2qy). \tag{31}$$

The Gram matrix is explicit:

$$\sum_q T_\chi(q)\overline{T_\eta(q)} = \begin{cases} p(p-2), & \chi = \eta, \\ -p, & \chi \neq \eta. \end{cases} \tag{32}$$

Indeed additive orthogonality forces the same y, while $y \mapsto (y+1)/y$ bijects $\mathbf{F}_p \setminus \{0, -1\}$ with $\mathbf{F}_p^\times \setminus \{1\}$. Equation (32) recovers the exact second moment

$$\sum_q |\widehat{1_A}(q)|^2 = p(K-1),$$

but the target (30) is an L^4 statement and therefore requires cancellation beyond this Gram identity.

Equivalently, solving $c(h) - c(k) = t$ gives

$$k = u_t(h) = \frac{(2-t)h + t}{2 + t - th}. \tag{33}$$

The matrix

$$U_t = \begin{pmatrix} 2-t & t \\ -t & 2+t \end{pmatrix}$$

has determinant 4 and trace 4. Therefore $\frac{1}{2}U_t$ lies in $\mathrm{SL}_2(\mathbf{F}_p)$, has trace 2, and is unipotent. In fact u_t is the Cayley conjugate of the translation $z \mapsto z - t$. Thus the same problem is an L^2 intersection problem between a multiplicative torus and a one-parameter unipotent family.

A second exact reformulation relates one Cayley sum fiber to one distinguished product class of a shifted subgroup. For $t \neq 2$ and $\lambda = t/(2-t)$,

$$((2-t)x + t)((2-t)y + t) = (2-t)P_t(x, y) + 4,$$

so on $P_t = 0$,

$$(x + \lambda)(y + \lambda) = (1 + \lambda)^2.$$

Known near-quadratic total multiplicative-energy estimates for shifted subgroups [15], and generic Möbius incidence bounds such as [16], are relevant but do not by themselves control the square-sum of this moving distinguished product class. This is the present pointwise gap.

Status: Open at the centered quadratic scale.. The pointwise $K^{8/3}$ theorem is exact; no theorem checked in the current audit upgrades it to (30).

12 A global split-prime budget for non-forced collision mass

The resultant structure gives a different kind of improvement when K is an odd prime: an exact average/sparsity theorem over all split rational primes. For $p \equiv 1 \pmod{K}$, define

$$\Delta_K(p) = \sum_{d \neq 0} r_{A-A}(d)^2 - (K-1)(2K-5).$$

This is exactly the number of collision quadruples not belonging to identity or central inversion.

Simultaneous multiplication of all four nonzero exponent indices by $a \in \mathbf{F}_K^\times$ acts freely on collision quadruples. The number of non-forced scaling orbits is

$$N_K = (K-1)(K-2)^2 - (2K-5) = K^3 - 5K^2 + 6K + 1. \tag{34}$$

Choose the unique representative of each orbit with first index $i = 1$, and write its collision polynomial and resultant as P_q and R_q.

Theorem 12.1 (Weighted split-prime collision budget)**.** *For every odd prime K,*

$$\sum_{p \equiv 1\,(K)} \Delta_K(p) \log p \leq N_K\big((K-1)\log 12 - 3\log K\big), \tag{35}$$

where the sum is over rational primes.

Proof. Fix a split prime p and a generator h of its order-K subgroup. For one normalized orbit q, its scaled member aq is a collision precisely when $P_q(h^a) = 0$. Thus the number $s_q(p)$ of colliding members of the orbit is the number of common roots of P_q and Φ_K among the $K-1$ primitive K-th roots in $\mathbf{F}_p$.

Because $p \neq K$, Φ_K is separable modulo p and splits completely. Hensel-lift its simple roots to $\mathbf{Z}_p$. Since Φ_K is monic,

$$R_q = \prod_{\Phi_K(\alpha)=0} P_q(\alpha),$$

so every common root contributes at least one factor of p and

$$s_q(p) \leq v_p(R_q).$$

On the other hand Theorem 6.2 gives $R_q = K^3 M_q^2$. A split prime is never K, hence $v_p(R_q) = 2v_p(M_q)$. Summing first over split primes and then over normalized orbits,

$$\begin{aligned}\sum_{p\equiv 1\,(K)} \Delta_K(p)\log p &= \sum_q \sum_{p\equiv 1\,(K)} s_q(p)\log p \\ &\le \sum_q 2\log|M_q| \\ &= \sum_q \log\frac{|R_q|}{K^3} \\ &\le N_K\big((K-1)\log 12 - 3\log K\big),\end{aligned}$$

using $|R_q| \le 12^{K-1}$. □

For every $P \ge 2$ and $L > 0$, Theorem 12.1 immediately gives

$$\sum_{\substack{p\equiv 1\,(K)\\ p\ge P}} \Delta_K(p) \le \frac{N_K C_K}{\log P}, \qquad \#\{p \ge P : p \equiv 1\ (K),\ \Delta_K(p) \ge L\} \le \frac{N_K C_K}{L\log P},$$

where $C_K = (K-1)\log 12 - 3\log K$. No prime number theorem in arithmetic progressions is used. A normalized average over a prescribed interval would need an independent lower bound for the number of split primes in that varying-modulus interval.

Status: Exact average/sparsity theorem.. Exact normalized resultant regressions for the complete $K = 11, 13$ exceptional sets verify the orbit/common-root/p-adic-valuation mechanism. They are finite checks, not the proof of Theorem 12.1.

13 The off/off elliptic route

After separating the mixed term, the unresolved contribution is the centered energy of $O(t)$. Historical symbolic work organizes the relevant fibers as a one-parameter genus-one family and suggests a quartic/Weierstrass model, a nonconstant rational j-map, visible bad fibers and same-j parameter symmetries.

Status: Theorem draft.. A publication-grade proof must re-establish or certify all of the following for the actual family:

1. a smooth projective model on an explicit open base;
2. the complete discriminant/bad-fiber locus, including characteristic exceptions;
3. affine/projective boundary corrections;
4. purity, lissity, geometric irreducibility, and uniform conductor/ramification control of the relevant R^1 sheaf;
5. all geometric isomorphisms and twists between parameter fibers;
6. separable and inseparable isogeny exclusions or explicit exceptional strata;
7. the exact hypotheses of the trace-sheaf correlation theorem used.

Michel-type family arguments [17], quantitative sheaf theory [18], Tate/isogeny technology [19], and the modern joint-family example of Grove–Saad [20] are relevant tools, but none is a plug-in proof for this pencil.

Conditional on the required square-root correlation with uniform constants, the existing combinatorics suggest

$$O\Big(K^2p + K^4p^{1/2} + K^4\Big)\,, \tag{36}$$

with intended regime $K = o(p^{1/4})$. Equation (36) remains a draft conclusion.

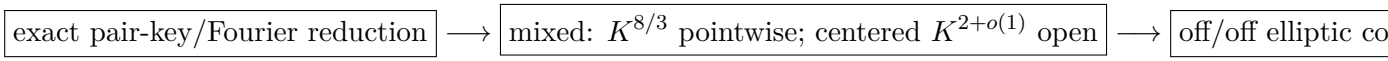
14 Finite spectral diagnostics and the microscopic BGS gap

The maintained finite spectral code diagonalizes the full complex Hermitian sector matrix

$$\mathcal{H}(q) = M(q) + M(q)^*.$$

A reconstruction-era implementation briefly replaced it by its entrywise real part; that bug was removed and guarded by regression. Current spacing records are descriptive empirical-CDF comparisons only. Pooled unfolded gaps are dependent, so no IID Kolmogorov–Smirnov significance is attached. The forward-window statistic retained in the repository is likewise not stationary number variance.

The logical chain is



The prime-budget theorem supplies an additional averaged control on exceptional mixed reductions; it does not silently replace the pointwise box. Closing an earlier box does not close a later one. A local-statistics theorem still requires an explicit comparison/local-law mechanism, or a genuinely arithmetic replacement, at mean-spacing scale with the correct symmetry class and uniformity.

15 Reproducibility and independent failure modes

The maintained verification stack is deliberately heterogeneous:

- Python/SymPy: transparent symbolic algebra, exact resultants, Galois-orbit/p-adic budget regressions, finite-field reference computations, primality and energy certificates;
- C++20: independent exact/exhaustive kernels, the direct Cayley-energy verifier, and the 28-digit Boost.Multiprecision Fibonacci witness;
- Lean 4: kernel-checked pair-key certificates, fixed Cayley controls, torus facts, historical falsifiers, and the bounded Fibonacci-ratio residual;
- TeX/CI: publication-facing integration and independent manuscript build.

The C++ arbitrary-precision verifier intentionally uses a different code path from Python. The Lean module does not formalize Carmichael's theorem; it certifies exactly the finite range left after the external primitive-divisor theorem is applied. This separation keeps the evidence graph honest.

The current tree contains no Julia sources. Historical Julia files remain accounted for in repository provenance; false algorithms were not ported merely to increase language diversity.

16 Conclusion

The exact pair-key/Fourier reduction and Cayley transform isolate the fourth-moment arithmetic obstruction sharply. Prime-order cotangent rigidity removes characteristic-zero accidental collisions. The identity

$$\mathrm{Res}(\Phi_K, P) = K^3 M^2$$

explains the systematic square structure seen in fixed-K computations. The Fibonacci slice shows why eliminating every exceptional prime is the wrong target: large collision norms and very large exceptional split primes need not produce large energy.

The direct bilinear-fiber theorem now closes a genuine pointwise polynomial-scale part of the mixed problem:

$$E_+(c(H \setminus \{1\})) < 34K^{8/3} \quad \text{for} \quad 100 < K < p^{3/4}/3.$$

The Galois-orbit resultant argument independently gives a global $O(K^4)$ logarithmic budget for all non-forced collision mass across split primes. What remains in the mixed term is a sharper centered-energy problem, not the earlier zero-collision problem. The off/off elliptic correlation remains a separate geometric theorem draft. Even success at the full moment level would still leave microscopic BGS as an independent local-statistics problem.

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